Summary: Monte Carlo Data Simulation Study

Data sets were generated and models were estimated in R (version 4.5.3) using brms (version 2.23.0) with the cmdstanr backend (CmdStan version 2.39.0). Priors were weakly informative and were not centered on the data-generating values; parameters shared by the correctly specified and misspecified models were assigned the same priors.

In the nonlinear exponential-decay models, the asymptote was assigned a Normal(0, 5) prior, the initial-elevation effect a Normal(0, 2) prior, and the decay rate a Normal(1, 1) prior truncated below at zero. In the linear models with a Day-1 indictor, the intercept and the Day-1 effect were assigned Normal(0, 5) and Normal(0, 2) priors, respectively.

In all models, the intercept of the log residual standard deviation was assigned a Normal(0, 1.5) prior. The subject-level random effects were parameterized by their standard deviations and their correlation matrix. Each random-effect standard deviation was assigned a half-Student-t prior with 3 degrees of freedom, location 0 and scale 1. The correlation matrix was assigned an LKJ(1) prior on its Cholesky factor.

Models under the Day-1 Scenario were first estimated with four Markov chain Monte Carlo (MCMC) chains of 5,000 iterations with 1,500 warm-up iterations, for a total of 14,000 draws. Models under the two scenarios that specified a decaying IE effect were first estimated with four MCMC chains of 8,000 iterations each, of which the first 2,000 were warm-up, resulting in 24,000 post-warm-up draws overall. No thinning was applied. All chains were initialized at zero on the unconstrained parameter scale, and sampling used adapt_delta = 0.98 and max_treedepth = 12. Convergence was assessed for all population-level parameters (fixed effects, random-effect standard deviations and random-effect correlations) using the rank-normalized split-$\hat{R}$ and the bulk and tail effective sample sizes (ESS; Vehtari et al., 2021). A fit was considered converged if the maximum $\hat{R}$ was ≤ 1.01, the minimum bulk and tail ESS were each ≥ 400, and no more than five divergent transitions occurred.

Replications that did not meet these criteria were specified to be refitted using a fixed sequence of increasingly conservative settings, each with a new random seed and max_treedepth = 15. For the day-1 scenario, refits were specified to use (Stage 1) 8,000 iterations with 2,000 warm-up iterations, and adapt_delta = 0.99, (Stage 2) 12,000 iterations with 4,000 warm-up iterations and adapt_delta = 0.995, and (Stage 3) 20,000 iterations with 6,000 warm-up iterations and adapt_delta = 0.995. For the other two scenarios involves a decaying IE effect, refits were specified as follows: (Stage 1) 10,000 iterations with 2,000 warm-up and adapt_delta = 0.99; (Stage 2) 16,000 iterations with 4,000 warm-up and adapt_delta = 0.995; and (Stage 3) 24,000 iterations with 6,000 warm-up and adapt_delta = 0.995. Refitting stopped at the first stage that met all criteria, and that fit was used for estimation; all reported convergence diagnostics refer to this final fit.

Across three scenarios, performance was evaluated for parameter bias, average posterior standard deviation, empirical standard deviation, root mean square error (RMSE), and 95% credible-interval coverage. Bias and RMSE were computed from posterior means. The sampling variability of the estimates across replications is denoted in the tables that follow by ‘SD of estimates’. The average standard error of an estimated is denoted in the tables that follow by ‘Average SE’.Following Collins, Schafer, and Kam (2001), we consider as problematic standardized bias values >0.4 and coverage below 90%.

Scenario 1: Misspecification of a fixed initial elevation at the first occasion as absent

Following the initial fit, 28 replications under the correctly specified model and 12 under the incorrectly specified model did not meet the convergence criteria and were refitted. After this first fit, 2 replications under the correctly specified model still did not meet the criteria and were refitted a second time (stage 2); all replications under the misspecified model met the criteria after the first refit. After refitting, all 250 replications under each model met all convergence criteria: the maximum rhat was 1.0098 under both correctly specified and the misspecified model, all bulk and tail ESS were >= 400, and no final fit had any divergent transitions.

Under this scenario, data were generated under a model in which the initial elevation was represented as a fixed effect at the first assessment. As shown in Table 1, the correctly specified model recovered the generating parameters well. Across parameters, (absolute) standardized bias was satisfactory (range = 0.24% to 15.36%), and empirical sampling variability was generally close to the average model-based standard error. Coverage of nominal 95% intervals ranged from 92.4% to 96.8%, indicating generally adequate parameter recovery and interval calibration.

Conversely, omitting the initial elevation from the fitted model produced bias in several parameters. The largest effects were observed for τ_0, the log of the residual standard deviation, which was estimated at -0.685 compared with the true value of -0.75 (standardized bias = 260.432%); the standard deviation of the random scale effect, which was underestimated (0.314 vs. 0.354; standardized bias = -204.544%); and the overall mean β_0, which was overestimated (2.554 vs 2.5; standardized bias = 108.590%). These biases corresponded to losses in coverage: empirical coverage was 21.6% for τ_0 and 57.2% for ϕ_c , compared with 92.4% and 96.8%, respectively, under the correctly specified model. Coverage for the mean response was 83.6% under the misspecified model versus 95.2% under the correctly specified model.

For this data-generating scenario, the results indicate that when a fixed initial elevation is confined to a single assessment, treating it as absent can distort estimation of the overall mean (overestimated), the fixed within-subject residual standard deviation (overestimated), the degree of between-subject heterogeneity of the level-1 residual standard deviation (underestimated).

Scenario 2: Misspecification of a fixed, exponentially decaying IE as confined to the first occasion

Following the initial fit, 14 replications under the correctly specified model and three under the misspecified model did not meet the convergence criteria and were refitted. After this first fit, all replications met the criteria (stage 2). Across all fittings with convergence criteria met, the maximum rhat was 1.00998, all bulk and tail ESS were >= 400, and no final fit had any divergent transitions.

In the second scenario, data were generated under a model in which IE was a fixed effect that decayed exponentially over time, and fitted models misspecified IE as a first-assessment-only effect or according to the correct exponential decay. As shown in Table 2, the correctly specified model recovered the generating parameters well. Across parameters, absolute standardized bias was small (range = 0.233% to 34.155%), and empirical sampling variability was generally close to the average model-based standard error. Coverage of nominal 95% intervals ranged from 92.4% to 97.6%, indicating generally adequate parameter recovery and interval calibration.

In contrast, specifying IE as an effect that impacted the first assessment alone produced bias in the overall mean, where the mean response was biased upward (2.53 vs. 2.5; standardized bias = 60.269%), and this corresponded to a loss in coverage, from 94.8% under the correctly specified model to 89.6% under the misspecified model. The magnitude of the initial IE effect was biased downward (0.474 vs. 0.5; standardized bias = -91.139%), and this corresponded to the greatest loss in coverage, from 94.8% under the correctly specified model to 83.2% under the misspecified model. Otherwise, the remaining model parameters were fairly well recovered and estimated coverage was satisfactory.

Scenario 3: Misspecification of a random, exponentially decaying IE as fixed

Following the initial fit, two replications under the correctly specified model and nine under the misspecified model did not meet the convergence criteria and were refitted. After this first fit, the two replications under the correctly specified model met the criteria (stage 2); all but two replications under the misspecified model met the criteria after the first refit. After refitting, the two replications under the misspecified model met all convergence criteria. Across all fittings after all convergence criteria were met for both models: the maximum rhat was 1.0098, all bulk and tail ESS were >= 400, and no final fit had any divergent transitions.

In the third scenario, data were generated under a model in which IE was a random, exponentially decaying effect, and fitted models specified IE as a fixed, or correctly as random. As shown in Table 3, the correctly specified model recovered the generating parameters well. Across parameters, absolute standardized bias was satisfactory (range = -3.463% to 24.172%), and empirical sampling variability was

generally close to the average model-based standard error. Coverage of nominal 95% intervals ranged from 92.0% to 96.8%, indicating generally adequate parameter recovery and interval calibration.

In contrast, specifying IE incorrectly as a fixed effect impacted the estimated initial elevation effect, as well as elements of the level-1 and level-2 covariance structures. The fixed initial elevation effect was biased upward (0.519 vs. 0.5; standardized bias = 42.475%), and this corresponded to a decrease in coverage, from 92.0% under the correctly specified model to 77.2% under the misspecified model. At the subject level, the estimated standard deviation of the random intercept was biased upward (0.854 vs. 0.837; standardized bias = 49.600%), and this corresponded to a slight, but acceptable, decrease in coverage, from 96.0% under the correctly specified model to 93.6% under the misspecifed model. The estimated standard deviation of the random scale was biased downward (0.328 vs. 0.354; standardized bias = -120.536%), and this corresponded to a decrease in coverage, from 94.8% under the correctly specified model to 80.8% under the misspecified model. At the occasion level, the estimated log residual standard deviation was biased upward (-0.691 vs. -0.75; standardized bias = 261.517%), and this corresponded to a decrease in coverage, from 95.6% under the correctly specified model to 28.4% under the misspecified model.

The remaining model parameters were well recovered and estimated coverage was satisfactory.

Table 1

Scenario 1: Misspecification of a fixed initial elevation at first occasion as absent

Model	Parameter	True Value	Average Estimate	Average SE	SD of Estimates	RMSE	% Coverage	% Standardized Bias
Correct	β_0	2.5	2.504	0.050	0.050	0.050	95.2	8.423
	β_1	0.5	0.500	0.027	0.028	0.028	95.2	-0.242
	ϕ_b	0.837	0.842	0.005	0.035	0.035	96.0	15.359
	ϕ_c	0.354	0.354	0.022	0.020	0.020	96.8	2.562
	ρ_{bc}	-.2	-.198	0.070	0.068	0.068	96.8	3.223
	τ_0	-0.75	-0.750	0.025	0.026	0.026	92.4	0.347
Misspecified	β_0	2.5	2.554	0.050	0.050	0.074	83.6	108.590
	ϕ_b	0.837	0.840	0.036	0.034	0.035	96.0	10.911
	ϕ_c	0.354	0.314	0.021	0.019	0.044	57.2	-204.544
	ρ_{bc}	-.2	-.197	0.073	0.071	0.071	94.0	4.397
	τ_0	-0.75	-0.685	0.023	0.025	0.070	21.6	260.432

Average Estimate = $\sum_1^R \hat{\theta}/R$. Average SE = $\sum_1^R \widehat{SE(\theta)}/R$. SD of estimates = SD($\hat{\theta}$). RMSE = root mean square error. % standardized bias = 100*[($\sum_1^R \hat{\theta}/R - \theta$)/ SD($\hat{\theta}$)]. Maximum MCSE bias was 0.004.

Table 2

Scenario 2: Misspecification of a fixed, exponentially decaying IE as confined to the first occasion

Model	Parameter	True Value	Average Estimate	Average SE	SD of Estimates	RMSE	% Coverage	% Standardized Bias
Correct	β_0	2.5	2.494	0.050	0.051	0.051	94.8	9.294
	β_1	0.5	0.500	0.028	0.028	0.028	94.8	-1.174
	λ	1.15	1.213	0.182	0.184	0.194	94.0	34.155
	ϕ_b	0.837	0.843	0.036	0.034	0.034	96.0	15.629
	ϕ_c	0.354	0.354	0.022	0.020	0.020	97.2	2.678
	ρ_{bc}	-.2	-.198	0.070	0.068	0.068	97.6	3.280
	τ_0	-0.75	-0.750	0.025	0.026	0.026	92.4	0.233
Misspecified	β_0	2.5	2.530	0.050	0.050	0.058	89.6	60.269
	β_1	0.5	0.474	0.028	0.028	0.038	83.2	-91.139
	ϕ_b	0.837	0.842	0.036	0.035	0.035	96.4	15.287
	ϕ_c	0.354	0.350	0.022	0.020	0.020	96.8	-19.503
	ρ_{bc}	-.2	-.198	0.070	0.068	0.067	96.8	3.595
	τ_0	-0.75	-0.743	0.025	0.026	0.027	92.0	26.298

Average Estimate = $\sum_1^R \hat{\theta}/R$. Average SE = $\sum_1^R \widehat{SE(\theta)}/R$. SD of estimates = SD($\hat{\theta}$). RMSE = root mean square error. % standardized bias = 100*[($\sum_1^R \hat{\theta}/R - \theta$)/ SD($\hat{\theta}$)]. Maximum MCSE bias was 0.012.

Table 3

Scenario 3: Misspecification of a random, exponentially decaying IE as fixed

Model	Parameter	True Value	Average Estimate	Average SE	SD of Estimates	RMSE	% Coverage	% Standardized Bias
Correct	β_0	2.5	2.494	0.050	0.049	0.049	92.8	-12.797
	β_1	0.5	0.498	0.043	0.045	0.045	92.0	-3.463
	λ	1.15	1.179	0.135	0.121	0.124	96.8	24.172
	ϕ_b	0.837	0.843	0.036	0.034	0.034	96.0	18.311
	ϕ_{IE}	0.5	0.496	0.043	0.046	0.046	94.8	-8.940
	ϕ_c	0.354	0.355	0.023	0.022	0.022	94.8	5.707
	ρ_{bc}	-.2	-.206	0.071	0.072	0.072	95.2	-8.838
	$\rho_{b,bIE}$	.15	.150	0.086	0.089	0.089	93.2	0.538
	$\rho_{bIE,c}$	-.1	-.090	0.108	0.107	0.107	94.8	9.077
	τ_0	-0.75	-0.748	0.026	0.025	0.025	95.6	7.605
Misspecified	β_0	2.5	2.491	0.051	0.049	0.050	92.4	-18.842
	β_1	0.5	0.519	0.031	0.045	0.049	77.2	42.475
	λ	1.15	1.205	0.186	0.170	0.178	97.6	32.040
	ϕ_b	0.837	0.854	0.037	0.035	0.039	93.6	49.600
	ϕ_c	0.354	0.328	0.021	0.021	0.033	80.8	-120.536
	ρ_{bc}	-.2	-.207	0.072	0.070	0.070	94.8	-10.206
	τ_0	-0.75	-0.691	0.024	0.023	0.064	28.4	261.517

Average Estimate = $\sum_1^R \hat{\theta}/R$. Average SE = $\sum_1^R \widehat{SE(\theta)}/R$. SD of estimates = SD($\hat{\theta}$). RMSE = root mean square error. % standardized bias = 100*[($\sum_1^R \hat{\theta}/R - \theta$)/ SD($\hat{\theta}$)]. Maximum MCSE bias was 0.011.